

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

Let $C : x^2 + y^2 = 4$ and $C' : x^2 + y^2 - 4\lambda x + 9 = 0$ be two circles. If the set of all values of λ so that the circles C and C' intersect at two distinct points, is $\mathbf{R} - [a, b]$, then the point $(8a + 12, 16b - 20)$ lies on the curve :

(1) $x^2 + 2y^2 - 5x + 6y = 3$

(2) $5x^2 - y = -11$

(3) $x^2 - 4y^2 = 7$

(4) $6x^2 + y^2 = 42$

Q2 - 2024 (01 Feb Shift 1)

Let the line $L : \sqrt{2}x + y = \alpha$ pass through the point of the intersection P (in the first quadrant) of the circle $x^2 + y^2 = 3$ and the parabola $x^2 = 2y$. Let the line L touch two circles C_1 and C_2 of equal radius $2\sqrt{3}$. If the centres Q_1 and Q_2 of the circles C_1 and C_2 lie on the y -axis, then the square of the area of the triangle PQ_1Q_2 is equal to

Q3 - 2024 (01 Feb Shift 2)

Let the locus of the mid points of the chords of circle $x^2 + (y - 1)^2 = 1$ drawn from the origin intersect the line $x + y = 1$ at P and Q . Then, the length of PQ is :

(1) $\frac{1}{\sqrt{2}}$

(2) $\sqrt{2}$

(3) $\frac{1}{2}$

(4) 1

Q4 - 2024 (27 Jan Shift 1)

Four distinct points $(2k, 3k)$, $(1, 0)$, $(0, 1)$ and $(0, 0)$ lie on a circle for k equal to :

(1) $\frac{2}{13}$

(2) $\frac{3}{13}$

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(3) $\frac{5}{13}$

(4) $\frac{1}{13}$

Q5 - 2024 (27 Jan Shift 2)

Consider a circle $(x - \alpha)^2 + (y - \beta)^2 = 50$, where $\alpha, \beta > 0$. If the circle touches the line $y + x = 0$ at the point P, whose distance from the origin is $4\sqrt{2}$, then $(\alpha + \beta)^2$ is equal to

Q6 - 2024 (29 Jan Shift 1)

Equation of two diameters of a circle are $2x - 3y = 5$ and $3x - 4y = 7$. The line joining the points $\left(-\frac{22}{7}, -4\right)$ and $\left(-\frac{1}{7}, 3\right)$ intersects the circle at only one point $P(\alpha, \beta)$. Then $17\beta - \alpha$ is equal to

Q7 - 2024 (30 Jan Shift 1)

If the circles $(x + 1)^2 + (y + 2)^2 = r^2$ and $x^2 + y^2 - 4x - 4y + 4 = 0$ intersect at exactly two distinct points, then

(1) $5 < r < 9$

(2) $0 < r < 7$

(3) $3 < r < 7$

(4) $\frac{1}{2} < r < 7$

Q8 - 2024 (30 Jan Shift 2)

Consider two circles $C_1: x^2 + y^2 = 25$ and $C_2: (x - \alpha)^2 + y^2 = 16$, where $\alpha \in (5, 9)$. Let the angle between the two radii (one to each circle) drawn from one of the intersection points of C_1 and C_2 be $\sin^{-1}\left(\frac{\sqrt{63}}{8}\right)$. If the length of common chord of C_1 and C_2 is β , then the value of $(\alpha\beta)^2$ equals

Q9 - 2024 (31 Jan Shift 1)

If one of the diameters of the circle $x^2 + y^2 - 10x + 4y + 13 = 0$ is a chord of another circle C , whose center is the point of intersection of the lines $2x + 3y = 12$ and $3x - 2y = 5$, then the radius of the circle C is

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(1) $\sqrt{20}$

(2) 4

(3) 6

(4) $3\sqrt{2}$

Q10 - 2024 (31 Jan Shift 2)

Let a variable line passing through the centre of the circle $x^2 + y^2 - 16x - 4y = 0$, meet the positive co-ordinate axes at the point A and B. Then the minimum value of $OA + OB$, where O is the origin, is equal to

(1) 12

(2) 18

(3) 20

(4) 24

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Questions with Answer Keys

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Answer Key

Q1 (4)		Q2 (72)		Q3 (1)		Q4 (3)	
Q5 (100)		Q6 (2)		Q7 (3)		Q8 (1575)	
Q9 (3)		Q10 (2)					

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Solutions

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Q1

$$x^2 + y^2 = 4$$

$$C(0, 0) \quad r_1 = 2$$

$$C'(2\lambda, 0) \quad r_2 = \sqrt{4\lambda^2 - 9}$$

$$|r_1 - r_2| < CC' < |r_1 + r_2|$$

$$|2 - \sqrt{4\lambda^2 - 9}| < |2\lambda| < 2 + \sqrt{4\lambda^2 - 9}$$

$$4 + 4\lambda^2 - 9 - 4\sqrt{4\lambda^2 - 9} < 4\lambda^2$$

$$\text{True } \lambda \in \mathbb{R} \dots (1)$$

$$4\lambda^2 < 4 + 4\lambda^2 - 9 + 4\sqrt{4\lambda^2 - 9}$$

$$5 < 4\sqrt{4\lambda^2 - 9} \text{ and } \lambda^2 \geq \frac{9}{4}$$

$$\frac{25}{16} < 4\lambda^2 - 9 \quad \lambda \in \left(-\infty, -\frac{3}{2}\right] \cup \left[\frac{3}{2}, \infty\right)$$

$$\frac{169}{64} < \lambda^2$$

$$\lambda \in \left(-\infty, -\frac{13}{8}\right) \cup \left(\frac{13}{8}, \infty\right)$$

from (1) and (2)

$$\lambda \in$$

$$\lambda \in \left(-\infty, -\frac{13}{8}\right) \cup \left(\frac{13}{8}, \infty\right) \Rightarrow \mathbb{R} - \left[-\frac{13}{8}, \frac{13}{8}\right]$$

$$\text{as per question } a = -\frac{13}{8} \text{ and } b = \frac{13}{8}$$

$\therefore$ required point is $(-1, 6)$ with satisfies option (4)

Q2

$$x^2 + y^2 = 3 \text{ and } x^2 = 2y$$

$$y^2 + 2y + 3 = 0 \Rightarrow (y + 3)(y + 1) = 0$$

$$y = -3 \text{ or } y = 1$$

$$y = 1, x = \sqrt{2} \Rightarrow P(\sqrt{2}, 1)$$

p lies on the line

$$\sqrt{2}x + y = \alpha$$

$$\sqrt{2}(\sqrt{2}) + 1 = \alpha$$

$$\alpha = 3$$

For circle C_1

Q_1 lies on y axis

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Solutions

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Let $Q_1(0, \alpha)$ coordinates

$$R_1 = 2\sqrt{3} \text{ (Given)}$$

Line L act as tangent

Apply $P = r$ (condition of tangency)

$$\Rightarrow \left| \frac{\alpha - 3}{\sqrt{3}} \right| = 2\sqrt{3}$$

$$\Rightarrow |\alpha - 3| = 6$$

$$\alpha - 3 = 6 \quad \text{or} \quad \alpha - 3 = -6$$

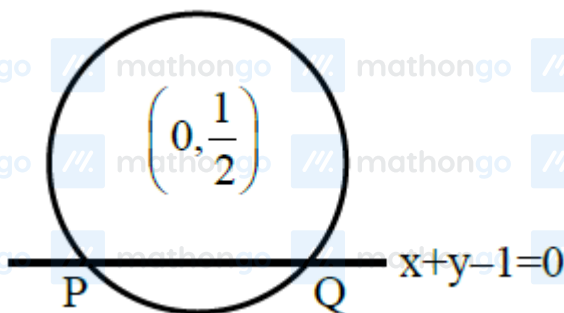
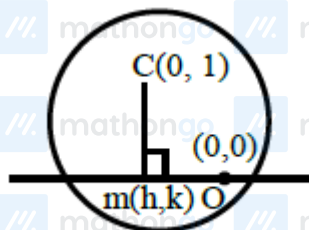
$$\Rightarrow \alpha = 9 \quad \alpha = -3$$

$$\Delta PQ_1Q_2 = \frac{1}{2} \begin{vmatrix} \sqrt{2} & 1 & 1 \\ 0 & 9 & 1 \\ 0 & -3 & 1 \end{vmatrix}$$

$$= \frac{1}{2} (\sqrt{2}(12)) = 6\sqrt{2}$$

$$(\Delta PQ_1Q_2)^2 = 72$$

Q3



$$m_{OM} \cdot m_{CM} = -1$$

$$\frac{k}{h} \cdot \frac{k-1}{h} = -1$$

$$\therefore \text{locus is } x^2 + y(y-1) = 0$$

$$x^2 + y^2 - y = 0$$

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Solutions

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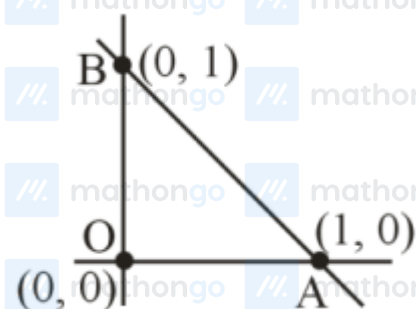
$$p = \left| \frac{1/2}{\sqrt{2}} \right| \quad p = \frac{1}{2\sqrt{2}}$$

$$PQ = 2\sqrt{r^2 - p^2}$$

$$= 2\sqrt{\frac{1}{4} - \frac{1}{8}} = \frac{1}{\sqrt{2}}$$

Q4

$(2k, 3k)$ will lie on circle whose diameter is AB.



$$(x-1)(x) + (y-1)(y) = 0$$

$$x^2 + y^2 - x - y = 0 \dots (i)$$

Satisfy $(2k, 3k)$ in (i)

$$(2k)^2 + (3k)^2 - 2k - 3k = 0$$

$$13k^2 - 5k = 0$$

$$k = 0, k = \frac{5}{13}$$

$$\text{hence } k = \frac{5}{13}$$

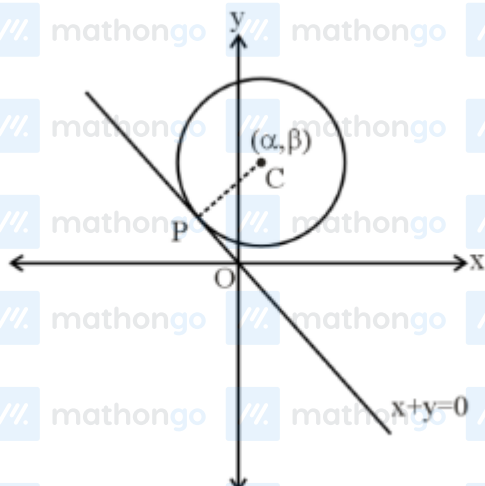
Q5

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Solutions

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$$S: (x - \alpha)^2 + (y - \beta)^2 = 50$$

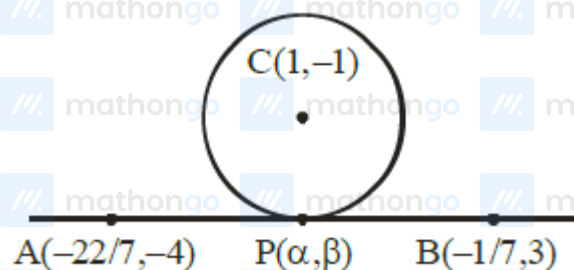
$$CP = r$$

$$\left| \frac{\alpha + \beta}{\sqrt{2}} \right| = 5\sqrt{2}$$

$$\Rightarrow (\alpha + \beta)^2 = 100$$

Q6

Centre of circle is (1, -1)



$$\text{Equation of AB is } 7x - 3y + 10 = 0 \dots (i)$$

$$\text{Equation of CP is } 3x + 7y + 4 = 0 \dots (ii)$$

Solving (i) and (ii)

$$\alpha = \frac{-41}{29}, \beta = \frac{1}{29} \therefore 17\beta - \alpha = 2$$

Q7

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Solutions

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If two circles intersect at two distinct points

$$\Rightarrow |r_1 - r_2| < C_1 C_2 < r_1 + r_2$$

$$|r - 2| < \sqrt{9 + 16} < r + 2$$

$$|r - 2| < 5 \text{ and } r + 2 > 5$$

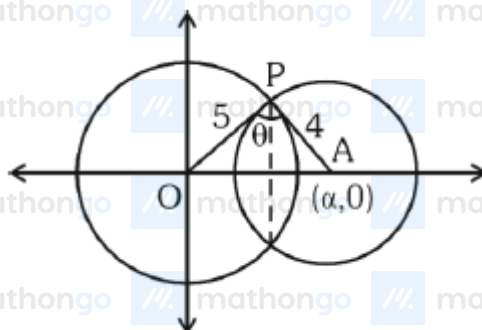
$$-5 < r - 2 < 5 \quad r > 3 \dots\dots(2)$$

$$-3 < r < 7 \dots\dots(1)$$

From (1) and (2)

$$3 < r < 7$$

Q8



$$C_1 : x^2 + y^2 = 25, C_2 : (x - \alpha)^2 + y^2 = 16$$

$$5 < \alpha < 9$$

$$\theta = \sin^{-1} \left(\frac{\sqrt{63}}{8} \right)$$

$$\sin \theta = \frac{\sqrt{63}}{8}$$

$$\text{Area of } \triangle OAP = \frac{1}{2} \times \alpha \left(\frac{\beta}{2} \right) = \frac{1}{2} \times 5 \times 4 \sin \theta$$

$$\Rightarrow \alpha \beta = 40 \times \frac{\sqrt{63}}{8}$$

$$\alpha \beta = 5 \times \sqrt{63}$$

$$(\alpha \beta)^2 = 25 \times 63 = 1575$$

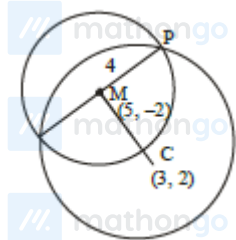
Q9

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Solutions

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$$2x + 3y = 12$$

$$3x - 2y = 5$$

$$13x = 39$$

$$x = 3, y = 2$$

Center of given circle is $(5, -2)$

$$\text{Radius } \sqrt{25 + 4 - 13} = 4$$

$$\therefore CM = \sqrt{4 + 16} = 5\sqrt{2}$$

$$\therefore CP = \sqrt{16 + 20} = 6$$

Q10

$$(y - 2) = m(x - 8)$$

$$\Rightarrow x\text{-intercept}$$

$$\Rightarrow \left(\frac{-2}{m} + 8 \right)$$

$$\Rightarrow y\text{-intercept}$$

$$\Rightarrow (-8m + 2)$$

$$\Rightarrow OA + OB = \frac{-2}{m} + 8 - 8m + 2$$

$$f'(m) = \frac{2}{m^2} - 8 = 0$$

$$\Rightarrow m^2 = \frac{1}{4}$$

$$\Rightarrow m = \frac{-1}{2}$$

$$\Rightarrow f\left(\frac{-1}{2}\right) = 18$$

$$\Rightarrow \text{Minimum} = 18$$

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